NV23 image172647 closeout report: aligned NV, T_2^*, and ^{13}C evidence

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Abstract

The project objective was to find a magnetic-field-aligned NV in the spatial range of SavedImage 3DXYZ-Image-2026-05-14-172647 and then obtain well-supported T_2^* and ^{13}C conclusions. The aligned branch is `reimage1804_c02`, tracked at roughly 39.7 kcps and accepted after terminal strong-pi pulsed ODMR showed a clear signal-only resonance near 3.87646 GHz. Ramsey data at three programmed detunings support only a short/few-microsecond T_2^* scale, not a robust scalar. The ^{13}C conclusion is stronger than Ramsey alone: det-shift Ramsey left a weak high-sideband-consistent candidate, and terminal CPMG N=8 independently showed a final-echo dip in the modeled $1/(4f_{13})$ target region. The supported final statement is: **reimage1804_c02 likely has a weak/moderate nearby-^{13}C-like coupling signature, but the current data do not support precise coupling extraction or a publication-grade single-spin claim.**

1 Final claims

- **Aligned NV:** established on `reimage1804_c02`. Strong-pi pODMR terminal raw/fitted-reference signal depth was 13.8–14.0%, centered at 3.876461010 GHz with approximately 0.69 MHz covariance uncertainty and 11.2 MHz FWHM. The reference did not show a matching depression (0.02%). Contrast is lower than the setup reference of about 22%, but is far above rejected `reimage1804_c01`.
- **T_2^*:** supported only as a short/few-μs order. Across terminal Ramsey fits, all-point fits were about 1.5–2.1 μs and fits excluding the first point were about 2.4–3.1 μs. Frequency/window/early-time handling changes the scalar, so the closeout value should be reported as *about 2–3* μs*, method-sensitive*, not as a precise final number.
- **^{13}C:** likely weak/moderate nearby-^{13}C-like coupling. Ramsey high-sideband target amplitudes followed the programmed detuning at about 1.95%, 0.84%, and 1.01% for detunings 1.50, 1.00, and 1.25 MHz, while fixed old high-frequency alternatives weakened. Terminal CPMG then showed a readout3/final-echo target-region dip at $\tau = 0.6736$ μs, 23.6 ns from the modeled $1/(4f_{13})$ target, with readout3/self=0.9528, readout3/readout1-baseline=0.9518, and readout3/readout2-baseline=0.9568.

2 Candidate search and acceptance path

The initial saved image was exported with an explicit `ImageData_YXZ`/kcps axis contract before any coordinates were used for tracking. The first candidate, `image172647_c01`, tracked once but later failed the pODMR execute before data because of low in-run counts; immediate standalone retrack also failed low-count. This was treated as count/freshness evidence rather than no-resonance evidence. A fresh re-image of the original region was therefore used.

`reimage1804_c01` tracked successfully but its terminal strong-pi pODMR produced only about 4–5% signal depressions against an expected approximately 22% contrast, so it was rejected for the aligned-NV branch. `reimage1804_c02` was the next ranked fresh candidate, tracked with final counts 39.690 kcps, and passed the alignment screen.

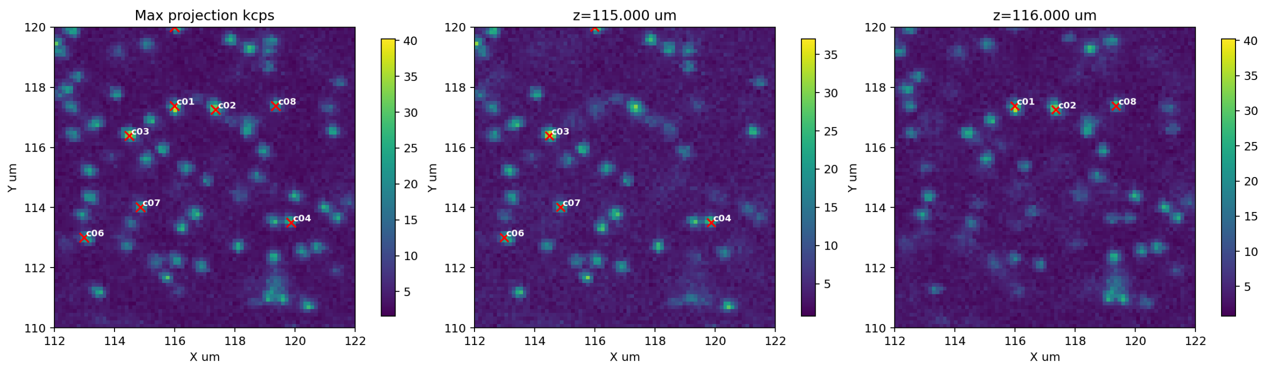

Figure 1: Fresh re-image candidate ranking used after the stale initial candidate failed low-count diagnostics. `reimage1804_c02` became the accepted aligned branch after `reimage1804_c01` was rejected by pODMR.

3 pODMR resonance evidence

The strong-pi pODMR alignment screen used `Rabimodulated.xml`, 3.825–3.925 GHz, 31 points, 4 averages, 50k repetitions/average, `mod_depth=1`, `length_rabi_pulse=52 ns`, `mw_ampl=-5`, `ampIQ=5`, and `freqIQ=50 MHz`. Terminal review verified readout roles: readout1 is the $m_S = 0$ reference and readout2 is the signal. The signal-only dip was strong enough to accept alignment, despite lower-than-reference contrast.

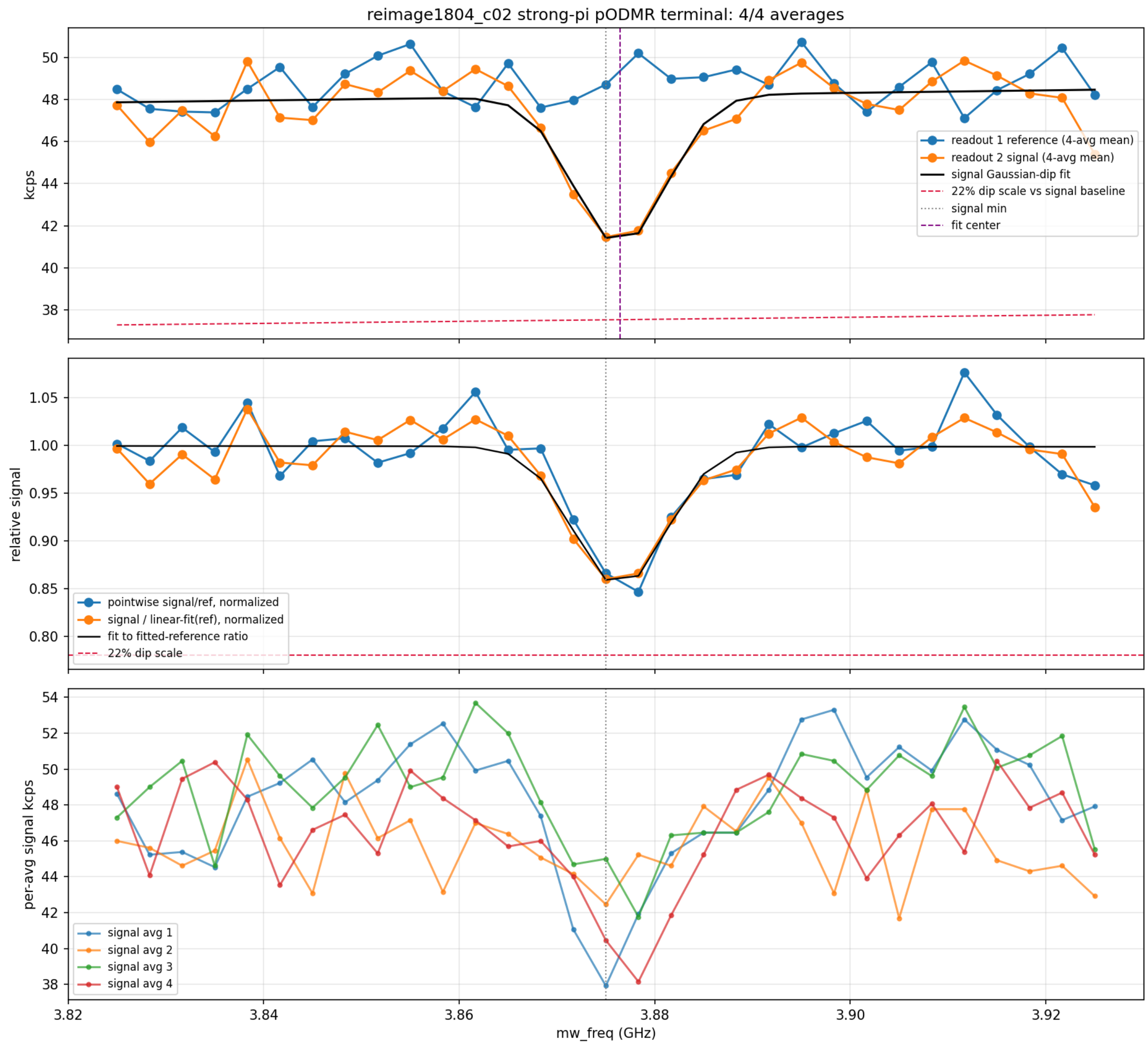

Figure 2: Terminal strong-pi pODMR for `reimage1804_c02`. The signal readout shows a clear dip near 3.87646 GHz without a matching reference dip.

A later weak-pi pODMR refinement used a narrower 6 MHz span around the strong-pi center. It refined the working microwave center to 3.876501337 GHz with about 47 kHz fit uncertainty, depth about 11.4%, and FWHM about 0.94 MHz. This frequency was used for later Ramsey/CPMG planning; pODMR by itself was not treated as ^{13}C evidence.

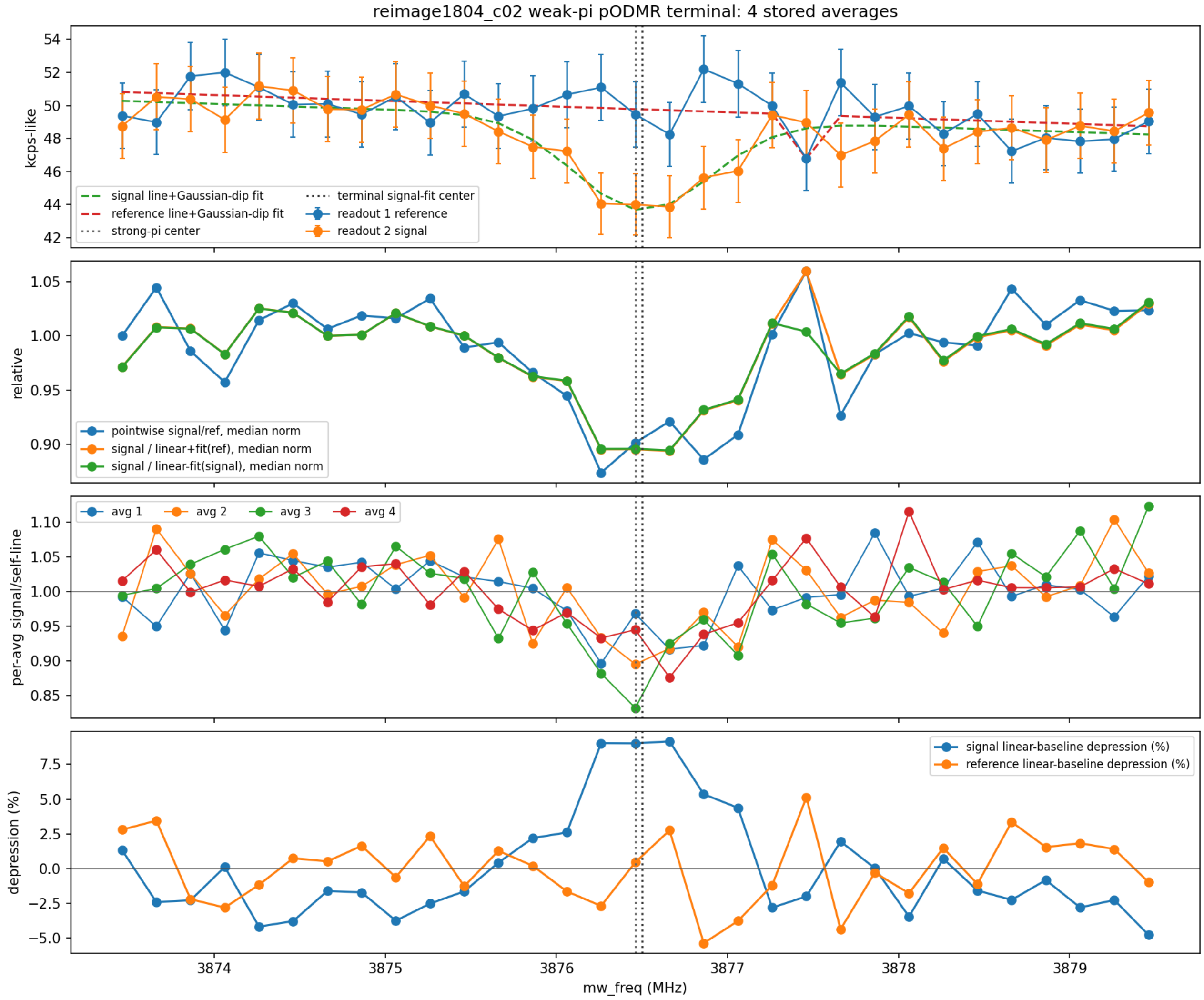

Figure 3: Terminal weak-pi pODMR resonance refinement. The refined center is close to the broad strong-pi center, ruling out a large electron-resonance mis-centering explanation for the earlier Ramsey high-frequency component.

4 Ramsey T_2^* and weak ^{13}C candidate

Three terminal Ramsey data sets were used. All used `ramsey.xml`, readout1 as the $m_S = 0$ reference and readout2 as the Ramsey signal, snake scan order, and no scan-order-aware drift flags. The det=1.5 MHz scout used the broad strong-pi center. The det=1.0 and det=1.25 MHz follow-ups used the weak-pi center and 800k planned shots/point.

Det (MHz)	Shots	τ span (μs)	FFT bin	direct f_{13}	det-f_{13}	carrier	det+f_{13}	T_2^* all	T_2^* excl. first
				readout2/self FFT amplitude (%)					fit, μs
1.50	8 x 50k	0–8	122 kHz	1.39	0.79	0.45	1.95	1.47	2.39
1.00	16 x 50k	0–10	100 kHz	0.14	0.73	0.41	0.84	2.11	2.94
1.25	16 x 50k	0–10	100 kHz	0.54	0.79	1.00	1.01	2.12	3.08

Table 1: Terminal Ramsey target-bin and fit summary. The high-sideband-compatible feature approximately tracks programmed detuning, but all target amplitudes are about-percent or sub-percent and the low-sideband/direct/carrier support is incomplete.

Figure 4: Terminal det=1.0 MHz Ramsey review. It supports short/few-μs T_2^* order and a weak det+f_{13} candidate, but not a standalone ^{13}C claim.

Figure 5: Terminal det=1.25 MHz Ramsey review. Static old-high/1.9 MHz alternatives are weak, and the strongest target bins are near the programmed carrier/high-sideband region, but the evidence remains low contrast.

5 CPMG N=8 independent discriminator

Because Ramsey left a weak but plausible det-shift-consistent component, a non-blind CPMG N=8 tau scan was designed from the weak-pi center. The expected ^{13}C Larmor frequency was about 384.6 kHz, giving modeled target conventions near $1/(4f_{13}) = 0.6500$ μs and $1/(2f_{13}) = 1.3001$ μs. The scan used `CPMG.xml`, $N = 8$, $\tau = 0.45$–1.60 μs, 37 points, 12 averages, 50k repetitions/average, and 600k shots/point. Terminal bridge status was clean: pre-run TrackCenter final 38.694 kcps, post-run final 44.535 kcps, no stop request, and no drift flags.

Exact nearest-grid points were not the best interpretation because the modeled filter-width scale was about $\tau/N \approx 81$ ns. A target-region review found the relevant minimum at $\tau = 0.6736$ μs, inside the $1/(4f_{13})$ target region. The feature appears in raw signal self-baseline and in reference-normalized diagnostics, so it is not a normalization-only artifact.

Averages	Shots/pt	readout3/self at 0.6736 μs	dip (%)	nearest 0.650 μs	nearest 1.300 μs
2	100k	0.9279	7.21	1.0197	0.9822
4	200k	0.9732	2.68	0.9908	0.9971
6	300k	0.9476	5.24	0.9850	0.9947
8	400k	0.9437	5.63	0.9983	0.9978
10	500k	0.9555	4.45	0.9998	0.9872
12	600k	0.9528	4.72	0.9871	0.9841

Table 2: Even-subset trend for the CPMG target-region point. The exact nearest precomputed 0.650 μs point understated the terminal feature; the region minimum at 0.6736 μs persisted in later balanced subsets.

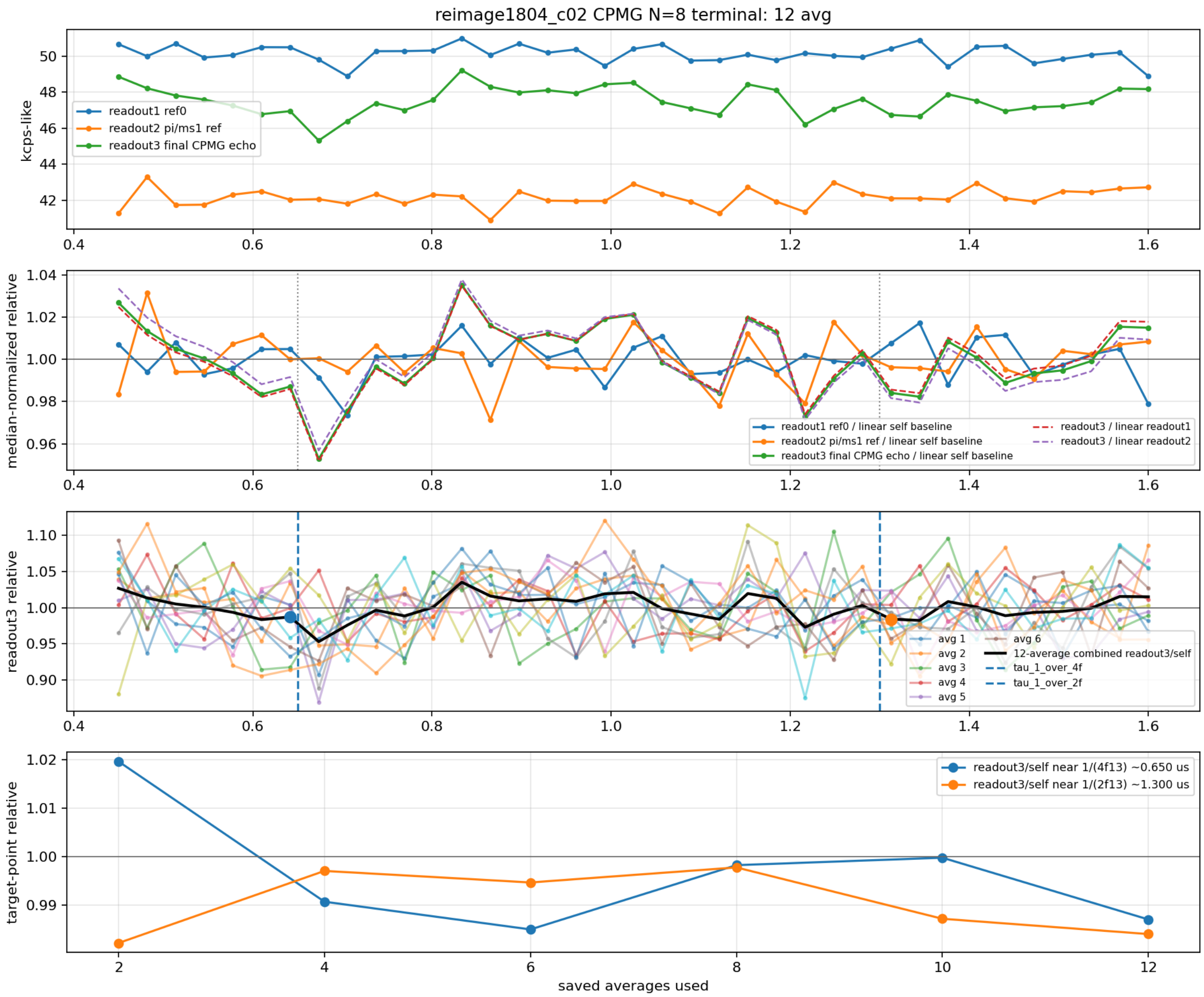

Figure 6: Terminal 12-average CPMG N=8 review over the full 0.45–1.60 μs scan.

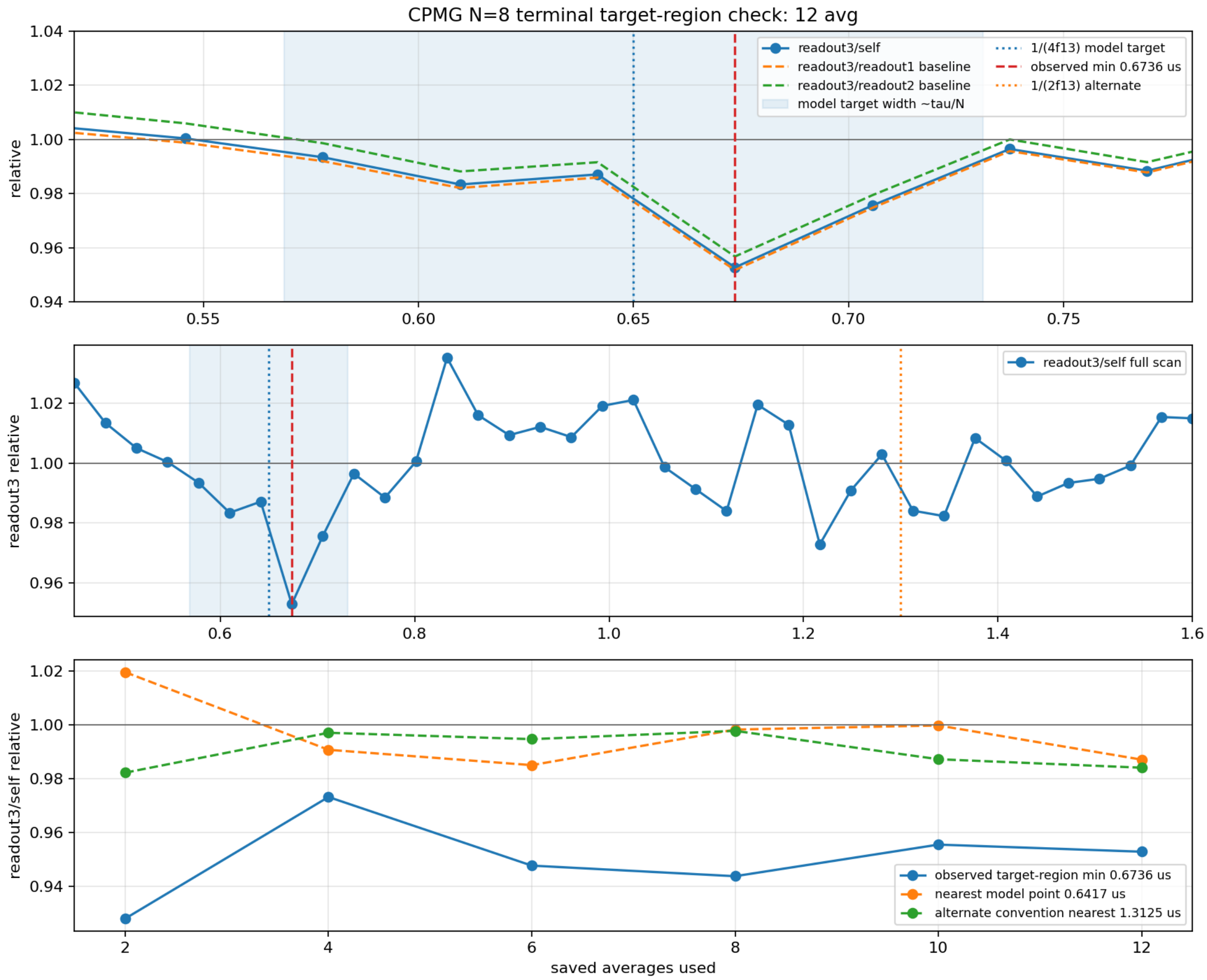

Figure 7: CPMG target-region review. The observed minimum at 0.6736 μs is within the modeled $1/(4f_{13})$ target region and corroborates the earlier weak det-shift Ramsey candidate.

6 Limitations and optional follow-up

- The strong-pi and weak-pi pODMR contrast on the accepted branch is lower than the approximately 22% setup contrast reference; this is a caveat on signal strength, not a rejection.
- The T_2^* scalar is sensitive to fit frequency, early-τ exclusion, and window choice. The reliable closeout is a short/few-μs order, not a precise scalar.
- The ^{13}C evidence is a supported weak/moderate nearby-^{13}C-like signature, not a precise coupling extraction. The CPMG target-region feature is the strongest corroborating evidence, but the project did not perform phase-resolved XY8/DDRF or a fine CPMG zoom.
- No further bridge-touching work is required for the original objective. If stronger confidence or parameter extraction is requested later, a fine CPMG scan around 0.60–0.72 μs or a phase-readout XY8/DDRF-style discriminator would be the next non-blind refinement, after fresh model, drift, verifier, and advisory gates.

7 Artifact index

Key project paths are in /home/takuya/.openclaw/workspace/.openclaw/projects/nv23_aligned_nv_t2star_13
The main evidence artifacts used here are:

- work/artifacts/analysis/reimage1804_c02_podmr_terminal_summary_4avg_20260514_1957.json
- work/artifacts/analysis/reimage1804_c02_weak_pi_podmr_terminal_4avg_summary_20260514_2251.js
- work/artifacts/analysis/reimage1804_c02_ramsey_terminal_8avg_summary_20260514_2150.json
- work/artifacts/analysis/reimage1804_c02_ramsey_det1_terminal_summary_16avg_20260515_0245.jso
- work/artifacts/analysis/reimage1804_c02_ramsey_det1p25_terminal_summary_16avg_20260515_0645.
- work/artifacts/analysis/reimage1804_c02_cpmg_n8_terminal_summary_12avg_20260515_0935.json
- work/artifacts/analysis/reimage1804_c02_cpmg_n8_terminal_target_region_review_20260515_0945.

8 Literature/prior-reference context

The report interpretation follows the project evidence and the current 23-C Siglent route, not old Tektronix timing assumptions. Literature/source checks used for context were: Doherty et al., "The nitrogen-vacancy colour centre in diamond," Physics Reports 528, 1–45 (2013), doi:10.1016/j.physrep.2013.02.001, available in the local literature index; and de Lange et al., "Universal dynamical decoupling of a single solid-state spin from a spin bath," Science 330, issue 6000 (2010), doi:10.1126/science.1192739, checked via Crossref on 2026-05-15. These references support the standard NV-center and dynamical-decoupling context; final numeric claims in this report come from the project artifacts above.